SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Computer Science and Engineering

CO1 ASSESSMENT TOOL 2 – Agile Sprint Planning & User Story Workshop

Course Code:	CSA10 / CSA1004	**Course Title:**	Software Engineering
CO Assessed:	CO1	**Assessment:**	Assessment Tool 2 – **Agile Sprint Planning & User Story Workshop**
Weightage:	20%	**Total Marks:**	25
CO1 Statement:	*Apply advanced methodologies and agile project management techniques to manage software projects effectively*		
Student Name:	AKSHAYA SEKAR	**Reg. No.:**	1911260204
Date:	15.10.2026	**Duration:**	60 minutes

Code of Conduct

I AKSHAYA SEKAR / 1911260204 (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: AKSHAYA SEKAR

Annexure A – Agile Sprint Planning & User Story Workshop

Instructions to Students:

Each student is assigned an individual software project problem statement. Based on the given problem statement, perform an **Agile Sprint Planning and User Story Workshop** by applying Agile software engineering practices. Prepare a comprehensive sprint planning document covering the following aspects:

Based on the given problem statement, perform an **Agile Sprint Planning and User Story Workshop** by applying Agile software engineering practices. Prepare a comprehensive sprint planning document covering the following aspects:

1. **Identify the Project Vision and Stakeholders**
   - Define the project's vision, objectives, and identify all key stakeholders involved.
2. **Create Epics and User Stories**
   - Break down the requirements into Epics and formulate well-structured User Stories using the format: *As a <user>, I want <feature>, so that <benefit>.*
3. **Define Acceptance Criteria**
   - Write clear and measurable acceptance criteria for each selected user story using the Given–When–Then format or equivalent.
4. **Prioritize the Product Backlog**
   - Organize and prioritize the product backlog based on business value, customer needs, and project dependencies.
5. **Plan the Sprint Backlog**
   - Select high-priority user stories for the first sprint and divide them into actionable development tasks.
6. **Estimate Effort Using Story Points**
   - Assign story points to each selected user story using an appropriate estimation technique (e.g., Fibonacci sequence or Planning Poker) and justify your estimates.
7. **Present the Sprint Goal and Expected Deliverables**
   - Define a clear sprint goal and describe the expected sprint deliverables, including the features to be completed at the end of the sprint.

PROJECT 32: INTELLIGENT HOTEL MANAGEMENT AND GUEST EXPERIENCE SYSTEM

1. PROJECT VISION AND STAKEHOLDER IDENTIFICATION

1.1 Project Vision

The vision of the Intelligent Hotel Management and Guest Experience System is to provide a single, integrated digital platform that helps a hotel manage its daily operations efficiently while delivering a smooth and personalized experience to guests. The system combines room inventory, reservations, check-in and check-out, room allocation, housekeeping, guest service requests, billing, guest histories, and performance analytics. By maintaining consistent information in one place, it aims to reduce booking conflicts, avoid duplicated work, improve response time, and support accurate decision-making.

The system is intended for use by small and medium-sized hotels as well as larger properties that need coordination between reception, housekeeping, accounts, and management. Intelligent features may include alerts for possible reservation conflicts, automatic room-status updates, prioritization of guest requests, and dashboards that summarize occupancy, revenue, and service performance. Staff remain responsible for confirming important decisions, correcting exceptional cases, and protecting guest information.

1.2 Project Objectives

- Maintain up-to-date room inventory, room types, rates, and availability.
- Create, modify, cancel, and track guest reservations.
- Support guest check-in, room allocation, and check-out.
- Record housekeeping assignments and room-cleaning status.
- Register guest requests and monitor their progress until completion.
- Generate itemized bills and invoices with accurate charges.
- Maintain guest histories for appropriate service personalization.

- Produce performance reports on occupancy, revenue, requests, and housekeeping.

1.3 Key Stakeholders and Their Roles

The main stakeholders are guests, reception staff, housekeeping staff, hotel managers, accounts staff, maintenance staff, system administrators, and the software development team. Guests use the system directly or through hotel staff to make reservations, provide preferences, and request services. Reception staff manage bookings, verify guest details, allocate rooms, and process check-in and check-out. Housekeeping staff receive room-cleaning tasks and update room status. Accounts staff prepare bills, record payments, and resolve billing questions. Maintenance staff respond to room faults and facility problems. Hotel managers review analytics and use them to plan staffing, pricing, and service improvements. System administrators manage user access, backups, and security, while developers test, maintain, and improve the application.

Each stakeholder has different access permissions. For example, housekeeping staff should see the room status and assigned tasks they need, but should not access payment-card details. Managers may view summarized operational reports, while guest records should only be available to authorized employees. This role-based access reduces accidental changes and protects confidential information.

2. EPICS AND USER STORIES

Epics represent broad areas of functionality. Each epic is divided into smaller user stories that can be developed, tested, and delivered during a sprint.

2.1 Epics

- Epic 1 – Room Inventory and Reservations: maintain room availability, rates, and booking records.
- Epic 2 – Guest Arrival and Departure: verify guest details, allocate rooms, and complete check-in/check-out.

- Epic 3 – Housekeeping and Maintenance: assign cleaning tasks and report rooms that need attention.
- Epic 4 – Guest Service Requests: record, assign, prioritize, and close guest requests.
- Epic 5 – Billing and Invoices: calculate room charges, service charges, taxes, and payments.
- Epic 6 – Guest History and Analytics: maintain appropriate visit history and generate operational reports.

2.2 User Stories

- US1 – As a receptionist, I want to view available rooms for selected dates, so that I can avoid double booking.
- US2 – As a guest, I want to make or modify a reservation, so that I can plan my stay conveniently.
- US3 – As a receptionist, I want to check in a guest and assign an available room, so that arrival is handled quickly.
- US4 – As a housekeeping staff member, I want to update a room as cleaning, cleaned, or inspected, so that reception sees the correct room status.
- US5 – As a guest, I want to submit a service request and view its status, so that I know whether it is being handled.
- US6 – As an accounts staff member, I want the system to calculate an itemized bill, so that charges are transparent and accurate.
- US7 – As a manager, I want to view occupancy and revenue reports, so that I can monitor hotel performance.

3. ACCEPTANCE CRITERIA

Acceptance criteria define the conditions that must be satisfied before a user story is considered complete. They are written in a testable Given–When–Then format.

- US1 – Room availability: Given valid arrival and departure dates, when the receptionist searches for rooms, then the system displays only rooms available for the full period and prevents overlapping confirmed reservations.
- US2 – Reservation: Given complete guest and date details, when a reservation is saved, then the system generates a unique booking reference and displays the reservation status. If required details are missing, it displays a clear validation message.
- US3 – Check-in: Given a confirmed reservation and an available, ready room, when authorized staff complete check-in, then the booking changes to checked-in and the room is linked to the guest.
- US4 – Housekeeping: Given a room-cleaning task, when staff update its status, then the latest status, assigned staff member, and update time are recorded and visible to authorized users.
- US5 – Guest request: Given a request with a category and description, when the guest or staff submits it, then the system creates a request number, assigns a status, and allows authorized staff to update it until closure.
- US6 – Billing: Given a guest stay and recorded services, when the bill is generated, then it lists room charges, additional services, applicable taxes, payments, and the final balance. The total must match the sum of the listed items.
- US7 – Analytics: Given valid operational records for a selected period, when a manager opens the dashboard, then the system displays occupancy, revenue, and request-status summaries with the selected date range.

4. PRODUCT BACKLOG PRIORITIZATION AND SPRINT PLANNING

4.1 Prioritized Product Backlog

The product backlog is ordered by business value, urgency, dependencies, and risk. The MoSCoW approach is used: Must Have items are essential for the first usable

release; Should Have items are important but can follow; Could Have items improve convenience; and Won't Have items are excluded from the current release.

ɔrity	Backlog Item	Reason	Planned Release
ıst Have	Room availability and reservation	Core booking function; prevents conflicts	Sprint 1
ıst Have	Guest details and check-in	Required to manage arrivals	Sprint 1/2
ıst Have	Room status and housekeeping	Prevents assignment of unready rooms	Sprint 2
ɔuld Have	Guest service request tracking	Improves response and accountability	Sprint 2
ɔuld Have	Billing and invoice generation	Supports accurate checkout and payment	Sprint 3
ıld Have	Guest history and preferences	Enables more personalized service	Sprint 3
ıld Have	Advanced analytics and forecasting	Supports planning after core data exists	Sprint 4
ɔn't Have now	Voice assistant and ernal smart-room integration	Optional and outside initial scope	Future release

4.2 Sprint 1 Backlog and Tasks

Sprint duration: 2 weeks. Sprint 1 focuses on a dependable room-availability and reservation workflow because later features depend on accurate booking records.

- Task 1: Confirm reservation fields, room types, rate rules, and booking status values.
- Task 2: Design the room and reservation database tables and their relationships.
- Task 3: Build the room inventory screen and date-based availability search.
- Task 4: Implement reservation creation, modification, cancellation, and unique booking references.
- Task 5: Add validation to prevent overlapping confirmed bookings for the same room.

- Task 6: Test normal bookings, invalid dates, unavailable rooms, cancellations, and simultaneous booking attempts.
- Task 7: Demonstrate the working flow and document defects and improvements.

5. STORY POINT ESTIMATION, SPRINT GOAL AND DELIVERABLES

5.1 Story Point Estimation

The team uses Planning Poker with a Fibonacci-like scale of 1, 2, 3, 5, 8, and 13 points. Story points represent relative effort, complexity, uncertainty, and testing needs rather than hours. The team discusses different estimates and agrees on a final value.

er Story	ints	tification
1 – Search room availability		quires date validation and iable availability rules.
2 – Create reservation		cludes guest details, booking reference, validation, and conflict evention.
sic room inventory setup		mple room records and status nagement.
servation test cases		vers standard, invalid, cancellation, and conflict enarios.

5.2 Sprint Goal

Sprint Goal: Develop and test the core room inventory and reservation features so that authorized hotel staff can check availability and create valid reservations without overlapping confirmed bookings.

5.3 Expected Deliverables

- A room inventory containing room numbers, room types, rates, and current status.
- A date-based room availability search screen.
- A reservation form with input validation and a unique booking reference.
- A rule that blocks overlapping confirmed reservations for the same room.

- Test results for successful bookings, invalid inputs, cancellations, and booking conflicts.
- A short demonstration and a list of known limitations or future improvements.

6. EXPECTED OUTCOMES, RISKS AND CONCLUSION

The completed system is expected to improve coordination between reception, housekeeping, accounts, and management. A shared source of information can reduce reservation conflicts, speed up guest service, improve housekeeping visibility, and make bills easier to verify. Guest histories and performance analytics can support better service and planning when used appropriately.

Key risks include unauthorized access to guest information, incorrect room-status updates, network or server downtime, inaccurate tax or rate settings, and staff resistance to a new workflow. These risks can be reduced through role-based access, secure authentication, regular backups, validation rules, audit logs, staff training, and systematic testing. Guest information should be collected only when necessary and retained according to the hotel's policies and applicable privacy requirements.

In conclusion, Agile development allows the hotel system to be delivered in small, testable increments. The first sprint concentrates on room inventory and reservations, creating a reliable foundation for check-in, housekeeping, guest requests, billing, guest history, and analytics in later sprints. Clear user stories, measurable acceptance criteria, prioritized backlog items, and realistic story-point estimates help the team deliver useful functionality while responding to feedback.

Rubrics for Calculation

Criteria	‹cellent (5 Marks)	›od (3 Marks)	ıtisfactory (2 Marks)	ɛeds Improvement (1 Mark)
›ject Vision & Stakeholder Identification (30%)	›ject vision is clear, comprehensive, and aligned with the problem statement. All relevant keholders and their roles are correctly identified.	›ion is clear with minor omissions. Most stakeholders are correctly identified.	›ion is partially defined. Some keholders are missing or incorrectly identified.	›ion is unclear or incomplete. keholders are poorly identified or missing.
›ics & User Stories (25%)	›ics are well-defined and user stories are complete, follow Agile format, and fully capture functional requirements.	›ics and user stories are mostly complete with minor formatting or requirement issues.	ɛr stories are partially complete with limited Agile structure.	ɛr stories are incomplete, unclear, or do not follow Agile practices.
›ceptance Criteria (20%)	›ceptance criteria are specific, measurable, testable, and aligned with all user stories.	›ceptance criteria are mostly clear and ›table with minor gaps.	›ceptance criteria are partially defined and lack clarity or completeness.	›ceptance criteria are missing or not aligned with user stories.
›oduct Backlog Prioritization & Sprint Planning (15%)	›oduct backlog is logically prioritized, sprint backlog is realistic, and task breakdown is complete.	›cklog prioritization and sprint planning are mostly appropriate with minor issues.	›oritization or sprint planning is partially complete with limited justification.	›cklog prioritization and sprint planning are poorly organized or incomplete.
›ry Point Estimation, Sprint Goal & Presentation (10%)	›ry point estimation is well justified, sprint goal is clear, deliverables are realistic, and presentation is professional and well organized.	›imation and sprint goal are mostly ›opriate with minor inconsistencies. ›sentation is clear.	›imation or sprint goal lacks justification. ›sentation is adequate but incomplete.	›imation is unrealistic or ›ssing. Sprint goal, deliverables, and presentation are unclear or incomplete.

Criteria	Mark
›ject Vision & Stakeholder Identification **(30%)**	/5
›ics & User Stories **(25%)**	/5
›ceptance Criteria **(20%)**	/5
›oduct Backlog Prioritization & Sprint Planning **(15%)**	/5
›ry Point Estimation, Sprint Goal & Presentation **(10%)**	/5
TOTAL	/25